MSG No. 147.14 $P\bar{3}1'$ [Type II, trigonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-3..1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `-3..1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `3..1'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,7,8,9]`
2	$[0, 0, -z]$	`[4,5,6,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `3..1'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[4,5,6,10,11,12]`

Table 5: Wyckoff site: `3e`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,10]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,9,12]`

Table 6: Wyckoff site: `3f`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,4,7,10]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,6,9,12]`

Table 7: Wyckoff site: `6g`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,7]`
2	$[-y, x-y, z]$	`[2,8]`

continued ...

Table 7

No.	position	mapping
3	$[-x+y,\ -x,\ z]$	[3,9]
4	$[-x,\ -y,\ -z]$	[4,10]
5	$[y,\ -x+y,\ -z]$	[5,11]
6	$[x-y,\ x,\ -z]$	[6,12]